

Percentage — 40 Difficult Questions

Questions only • No answers or solutions

1. Q274 — *Successive decrease*

A person's salary is decreased by steps of 20%, 15% and 10%. Approximately by what percent should the reduced salary be increased so as to get back the original salary? (a) 39% (b) 44% (c) 56% (d) 63%

2. Q276 — *Numerator and denominator*

If the numerator of a fraction is increased by 240% and the denominator of the fraction is decreased by 50%, the resultant fraction is $\frac{5}{6}$. What is the original fraction? (a) $\frac{1}{4}$ (b) $\frac{2}{3}$ (c) $\frac{5}{12}$ (d) $\frac{4}{11}$ (e) None of these

3. Q282 — *Price increase and expenditure*

The price of sugar increases by 32%. A family reduces its consumption so that the expenditure on sugar is up only by 10%. If the total consumption of sugar before the price rise was 10 kg per month, then the consumption of sugar per month at present is (a) 8 kg (b) $8\frac{1}{2}$ kg (c) $8\frac{3}{4}$ kg (d) 9 kg

4. Q375 — *Election*

In a village 60% votes were cast in an election. A and B were the contestants. A won by 600 votes. If B had got 40% more votes, there would have been a tie between them. Find the number of recognized voters in the village. (a) 4500 (b) 2800 (c) 3500 (d) 3600

5. Q157 — *Election, non-voters and invalid votes*

10% of the voters did not cast their vote in an election between two candidates. 10% of the votes polled were found invalid. The successful candidate got 54% of the valid votes and won by a majority of 1620 votes. The number of voters enrolled on the voters' list was (a) 25000 (b) 33000 (c) 35000 (d) 40000

6. Q164 — *Eligible candidates*

In an examination, 5% of the applicants were found ineligible and 85% of the eligible candidates belonged to the general category. If 4275 eligible candidates belonged to other categories, then how many candidates applied for the examination? (a) 30,000 (b) 35,000 (c) 37,000 (d) None of these

7. Q165 — *Marks*

Two students appeared at an examination. One of them secured 9 marks more than the other and his marks was 56% of the sum of their marks. The marks obtained by them are: (a) 39, 30 (b) 41, 32 (c) 42, 33 (d) 43, 34

8. Q46 — *Passing percentages*

In an examination, 80% of the students passed in English, 85% in Mathematics and 75% in both English and Mathematics. If 40 students failed in both the subjects, find the total number of students.

9. Q277 — *Price increase and consumption reduction*

The price of a certain item is increased by 15%. If a consumer wants to keep his expenditure on the item same as before, how much percent must he reduce his consumption of that item? (a) $20\frac{10}{23}\%$ (b) $13\frac{1}{23}\%$ (c) $16\frac{2}{3}\%$ (d) 15%

10. Q278 — Oil price

If the price of oil is increased by 30%, then by how much percent a family should reduce its consumption so that the expenditure would remain the same? (a) $15\frac{1}{23}\%$ (b) $15\frac{3}{14}\%$ (c) $23\frac{1}{13}\%$ (d) $12\frac{76}{13}\%$

11. Q279 — Price fall

The price of wheat falls by 16%. By what percentage a person can increase the consumption of wheat so that his overall budget does not change? (a) 16% (b) 18% (c) 18.5% (d) 19%

12. Q280 — Ratio

The price of oil is increased by 25%. If the expenditure is not allowed to increase, the ratio between the reduction in consumption and the original consumption is (a) 1:3 (b) 1:4 (c) 1:5 (d) 1:6

13. Q281 — Sugar price

The price of sugar per kg increased from ₹16 to ₹20. The percentage reduction in the use of sugar so that the expenditure does not increase, should be (a) 15% (b) 20% (c) 25% (d) 40%

14. Q350 — Sugar price reduction

The price of sugar having gone down by 10%, Sharad can buy 6.2 kg more for ₹279. The difference between the original and the reduced price (per kg) is (a) ₹0.50 (b) ₹1 (c) ₹1.50 (d) ₹4.50

15. Q351 — Sugar price fall/rise

If the price of sugar falls by 12.5%, a person can buy 9 kg more of sugar for ₹1260 than before. If the price had risen by 12.5%, how much sugar would he have bought for the same sum? (a) 288 kg (b) 312 kg (c) 328 kg (d) 336 kg

16. Q353 — Two-subject failure

In an examination, 34% of the students failed in Mathematics and 42% failed in English. If 20% of the students failed in both the subjects, then the percentage of students who passed in both the subjects was (a) 44 (b) 50 (c) 54 (d) 56

17. Q354 — Newspapers

40% of the people read newspaper X, 50% read newspaper Y and 10% read both the papers. What percentage of the people read neither newspaper? (a) 10% (b) 15% (c) 20% (d) 25%

18. Q355 — Football and cricket

Out of 450 students of a school, 325 play football, 175 play cricket and 50 neither play football nor cricket. How many students play both football and cricket? (a) 50 (b) 75 (c) 100 (d) 225

19. Q356 — Two types of lunch

In a hotel, 60% had vegetarian lunch while 30% had non-vegetarian lunch and 15% had both types of lunch. If 96 people were present, how many did not eat either type of lunch? (a) 20 (b) 24 (c) 26 (d) 28

20. Q357 — Hockey and football

There are 600 boys in a hostel. Each plays either hockey or football or both. If 75% play hockey and 45% play football, how many play both? (a) 48 (b) 60 (c) 80 (d) 120

21. Q358 — Tea and coffee

In a certain office, 72% of the workers prefer tea and 44% prefer coffee. If each of them prefers tea or coffee and 40 like both, the total number of workers in the office is (a) 200 (b) 240 (c) 250 (d) 320

22. Q359 — Two-subject failure

In an examination, 30% and 35% students respectively failed in History and Geography while 27% students failed in both the subjects. If the number of students passing the examination is 248, find the total number of students who appeared in the examination.

23. Q148 — Different passing percentages

In a test, minimum passing percentage for girls and boys is 35% and 40% respectively. A boy scored 483 marks and failed by 117 marks. What is the minimum passing marks for girls? (a) 425 (b) 500 (c) 520 (d) 625 (e) None of these

24. Q149 — Failed by percentage

In an examination it is required to get 40% of the aggregate marks to pass. A student got 261 marks and is declared failed by 4% marks. What are the maximum aggregate marks a student can get? (a) 700 (b) 730 (c) 745 (d) 765 (e) None of these

25. Q151 — Passing and obtained marks

In an examination it is required to get 296 of the total maximum aggregate marks to pass. A student gets 259 marks and is decided failed. The difference of marks obtained by the student and that required to pass is 5%. What are the maximum aggregate marks a student can get? (a) 690 (b) 740 (c) 780 (d) Cannot be determined

26. Q152 — Election margin

Two candidates fought an election. One of them got 62% of the total votes and won by 432 votes. What is the total number of votes polled? (a) 1500 (b) 1600 (c) 1800 (d) Cannot be determined (e) None of these

27. Q153 — Invalid votes

In a college election between two candidates, one candidate got 55% of the total valid votes. 15% of the votes were invalid. If the total votes were 15200, what is the number of valid votes the other candidate got? (a) 5814 (b) 6840 (c) 7106 (d) 8360 (e) None of these

28. Q154 — Invalid votes and margin

At an election involving two candidates, 68 votes were declared invalid. The winning candidate secures 52% and wins by 98 votes. The total number of votes polled is (a) 2382 (b) 2450 (c) 2518 (d) None of these

29. Q155 — Votes cast

In a certain assembly constituency election, 80% of voters exercised their voting right and the winning candidate got elected with 65% of votes polled. What percent of total votes did he poll? (a) 35 (b) 52 (c) 55 (d) 57

30. Q156 — Invalid vote percentage

In an election, a total of 5,00,000 voters participated. A candidate got 2,55,000 votes which was 60% of the total valid votes. What was the percentage of invalid votes? (a) 10% (b) 12% (c) 15% (d) 300/17%

31. Q160 — Weighted sales tax

While purchasing one item costing ■400, I had to pay the sales tax at 7% and on another costing ■6400, the sales tax was 9%. What percent of the sales tax I had to pay, taking the two items together on an average? (a) 8% (b) $13\frac{8}{17}\%$ (c) $15\frac{8}{17}\%$ (d) $8\frac{1}{2}\%$

32. Q161 — Aggregate percentage

A student secures 90%, 60% and 54% marks in test papers with 100, 150 and 200 respectively as maximum marks. The percentage of his aggregate is (a) 64 (b) 68 (c) 70 (d) None of these

33. Q162 — Boys and girls

1100 boys and 700 girls are examined in a test; 42% of the boys and 30% of the girls pass. The percentage of the total who failed is (a) 58% (b) $62\frac{2}{3}\%$ (c) 64% (d) 78%

34. Q163 — Age distribution

In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is $\frac{2}{3}$ of the number of students of 8 years age which is 48. What is the total number of students in the school? (a) 72 (b) 80 (c) 120 (d) 150 (e) None of these

35. Q45 — Two-subject failure

In an examination, 35% of total students failed in Hindi, 45% failed in English and 20% in both. Find the percentage of those who passed in both the subjects.

36. Q39 — Population

During one year, the population of a town increased by 5% and during the next year, the population decreased by 5%. If the total population is 9975 at the end of the second year, then what was the population size in the beginning of the first year?

37. Q40 — Reverse percentage

If A earns 33■% more than B, how much percent does B earn less than A?

38. Q41 — Salary comparison

If A's salary is 30% less than that of B, then how much percent is B's salary more than that of A?

39. Q43 — Evaporation

After 25 kg of water had been evaporated from a solution of salt and water, which had 20% salt, the remaining solution had 30% salt. Find the weight of the original solution.